Five Safes Workflow RO-Crate and WfExS

Closing the gap of federated analysis and Trusted Research Enviroments (TREs) in the Health Data context

Eugenio Gonzalo[1,2], Laia Codó[1,2], Jose María Fernández[1,2], Stian Soiland-Reyes[3,4], Salvador Capella-Gutierrez[1,2], Emily Jefferson[5], Carole Goble[3], Tim Beck[6], Phil Quinlan[6], Tom Giles[6]

ELIXIR All Hands , 10-12 June 2024, Uppsala, Sweden

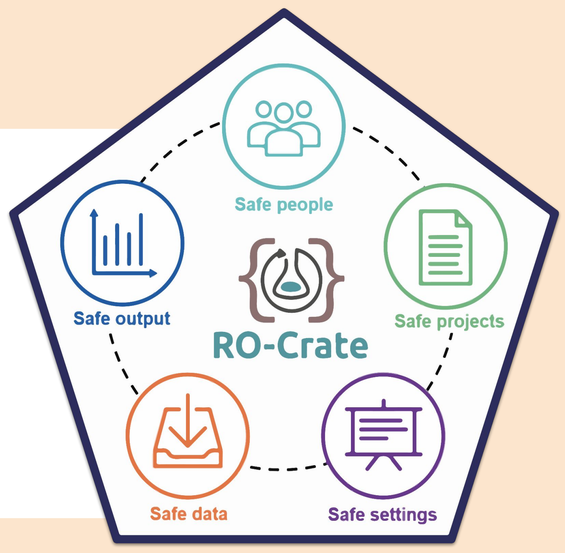
Towards a network of **Trusted Research Environments** for sensitive health and socio-economic data

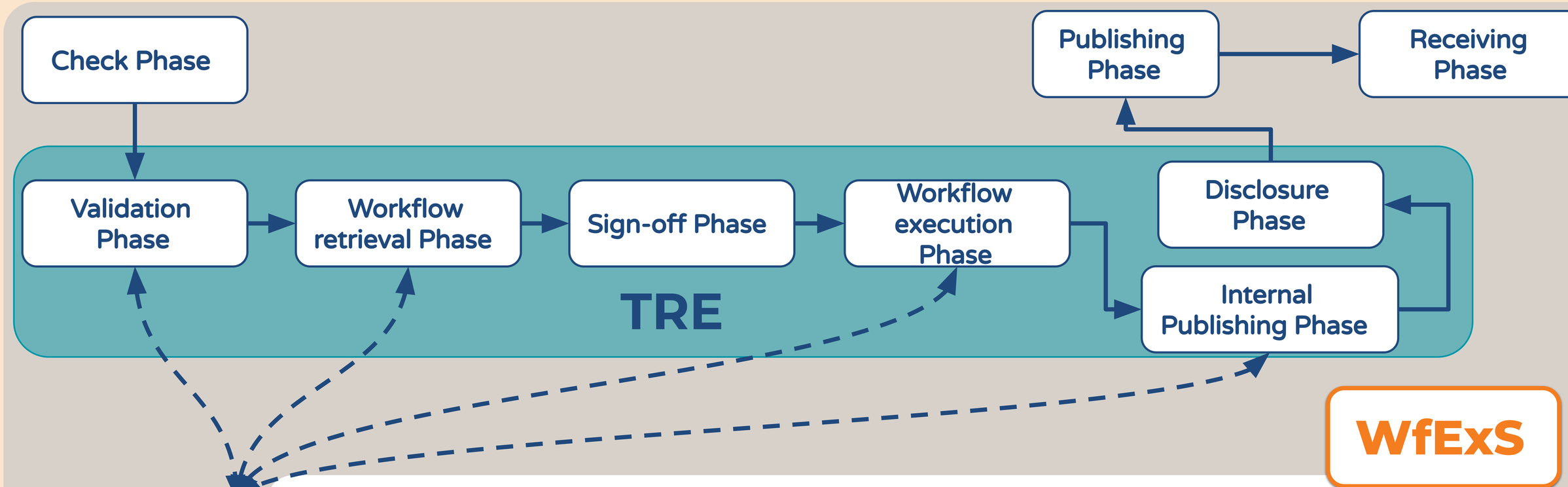
eosc | ENTRUST
European Network of Trusted Research Environments

A Trusted Research Environment (TRE) provides approved researchers from trusted organisations with timely and secure access to health-related data for secondary use, e.g. research purposes. It also provides a secure data platform with the analytical and statistical tools to support researchers' work. The need to manually check analysis results represents the major bottleneck for their up-scale and broad adoption.

Five Safes + RO-Crate → **Five Safes RO-Crate**

Five Safes RO-Crate represents a unit of computational workflow-based access to sensitive information which is managed in accordance with a set of principles conforming to the Five Safe framework. It extends the Workflow Run RO-Crate for workflow analysis provenance to carry metadata for procedures operating the 5-safes components: data, people, projects, settings and outputs.





WfExS

WfExS is a high-level workflow orchestrator that has been proposed to play that role in a TRE. It is workflow agnostic, delegating execution to workflow managers (Nextflow, CWL), and it can ingest and produce RO-Crates within restrictive environments. To offer a reproducible execution, the orchestrator prepares a secure and isolated working environment, materialising data and software beforehand; all the gathered details and provenance are reflected in the output RO-Crate metadata.

Learn more: Five Safes | RO-Crate | Five Safes RO-Crate | WfExS

Contact
Eugenio Gonzalo Jimenez
eugenio.gonzalo@bsc.es

1 Spanish National Bioinformatics Institute (INB/ELIXIR-ES)
2 Barcelona Supercomputing Center (BSC)
3 The University of Manchester (UNIMAN)
4 University of Amsterdam (UvA)
5 HDR UK
6 University of Nottingham (UNOTT)

EOSC-ENTRUST project receives funding from the European Union's Horizon Europe Programme under grant agreement number 101131056

INB Spanish National Bioinformatics Institute
eosc | ENTRUST European Network of Trusted Research Environments
BSC Barcelona Supercomputing Center Centro Nacional de Supercomputación
Funded by the European Union
ELIXIR ALL HANDS 2024
/company/elixir-europe